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How a Tailored Intervention Workflow for Large Size Completion Averted a Remedial Workover Rig Operation

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Abstract

Despite higher cost implications, it is not uncommon that fishing work in large size completions ends up conducted by a workover rig instead of traditional rigless intervention methods. Top of fish (TOF) accessibility, alignment between the fishing tool and the fish, and restricted pulling capacity are some of the typical challenges encountered around this work. This paper describes a tailored intervention workflow for coiled tubing (CT) fishing operation in large size completion that enabled averting a costly workover rig operation.

The objective of the intervention was the fishing and retrieval of a wellbore isolation plug stuck inside a 9 5/8-in. × 7-in. completion. Initial fishing attempts with slickline were unsuccessful as TOF was not accessible due to the presence of debris. Extensive CT software modeling was carried out for guided selection of the CT pipe, fishing tools, and modeling of cleanout work. TOF accessibility was restored through wellbore cleanout with high-pressure rotary jetting tool. Then, the centralized fishing assembly equipped with flow-release fishing tool, allowed optimal tool manipulation and effective latching of the fish neck. Upon applying the preset pulling force, the isolation plug was unset and retrieved to the surface.

The rigless fishing intervention was successfully conducted in two CT runs. The initial run consisted of a wellbore cleanout to restore accessibility to the fish. In line with CT hydraulics modeling, cleanout operation was carried out with nitrified fluids to deliver the required annular velocity for efficient solids transportation inside the 9 5/8-in. casing. Likewise, high-pressure rotary jetting tool was used for 360-degree wellbore coverage and effective scale removal. Accessibility to the TOF was restored at the end of this stage, and close to 150 kg of scale were recovered at surface. The fishing operation, executed in the second run, achieved first-attempt success with a 34,000-lbs overpull, retrieving the plug without damage. Centralization requirements across the transition from 9 5/8-in. to 7-in. casing were resolved through dynamic adjustments in the fishing assembly. Extensive pre-job planning, including detailed simulations and risk assessments, played a crucial role in this success. The tailored intervention eliminated the need for workover rig deployment, restored the well to production in less than six days, and reduced operational costs. These results underscore the operational robustness and flexibility of CT in complex fishing scenarios.

As the industry strives for cost-effective and sustainable interventions, the case study presented in this paper provides an example of how a custom-fit CT intervention workflow for complex fishing operation saved a costly workover rig intervention, while reducing near 70% CO₂ emissions.

Introduction

The subject well is a water injector completed with a 9 5/8-in. casing and a 7-in. liner, terminating in a 6 1/8-in. open hole. The well was in continuous operation for over five years before being temporarily taken offline. To facilitate a surface wellhead repair, the well was killed and isolated using a 6-in. mechanical bridge plug deployed via slickline within the 7-in. liner. Following completion of the wellhead repair, an attempt was made to retrieve the mechanical plug using slickline operations. However, the retrieval was unsuccessful due to significant debris accumulation on top of the plug, which prevented proper engagement with the fishneck (Fig. 1).

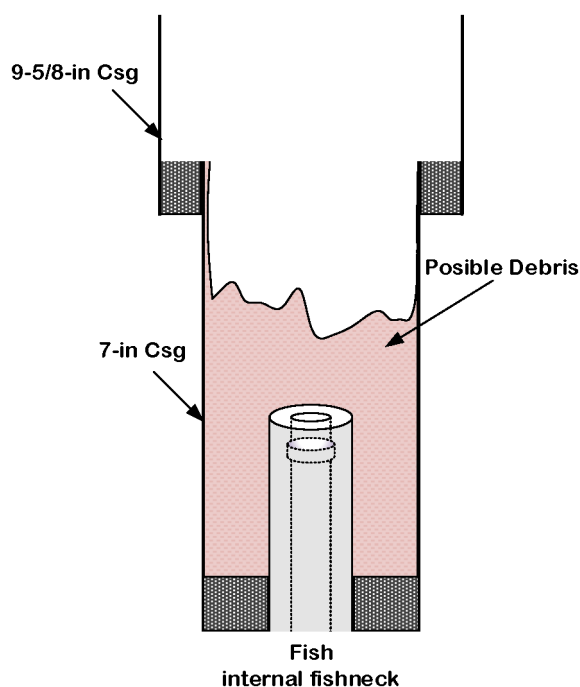


Figure 1—Well schematic and representation of the fish.

As a result, coiled tubing (CT) intervention was required to perform a cleanout and re-establish access to the plug for successful retrieval. Given that the well is a water injector, the production flowline is equipped with check valves designed to permit unidirectional flow toward the reservoir and prevent any backflow to surface facilities. As a result, a dedicated surface well testing package was required to bypass the injection flowline and manage the produced fluids during the operation. The surface package included a choke manifold, a sand filter, and a desander unit, all configured to effectively process and contain the return fluids and solids generated during the cleanout. It is noteworthy that similar well interventions in the field had faced considerable challenges—primarily due to complex wellbore geometries and debris buildup—which led several service providers to decline further operations on comparable well profiles after unsuccessful attempts.

Following a comprehensive risk assessment and collaborative discussions, the plug provider advised against proceeding with the operation using CT, citing a success probability of less than 5% and recommending a workover rig instead. However, a detailed proposal with a specific design to mitigate the

identified risks and significantly enhance the likelihood of the CT operational success was prepared for this intervention.

Solution

Hydraulic Simulations

A comprehensive engineering analysis was conducted, to simulate the cleanout and fishing operation with CT. The initial hydraulic simulations revealed that performing the cleanout using lean fluid at a pumping rate of 2.5 bbl/min resulted in insufficient annular velocities—particularly in the 9 5/8-in. casing section, where the velocity dropped to approximately 35 ft/min. This is significantly below the recommended threshold of 80 to 60 ft/min required for effective cleanout with gelled fluids. To meet these velocity requirements, a liquid rate of at least 5.0 bbl/min and the use of gelled fluid would be necessary to ensure operational success.

To overcome these limitations, an alternative approach was evaluated using a two-phase flow system. By introducing nitrogen at a rate of 800 scf/min in conjunction with a 2.5 bbl/min liquid rate, the modeled annular velocities improved substantially, reaching 125 to 170 ft/min in the 7-in. section and 80 to 190 ft/min in the 9 5/8-in. section. As presented in [Table 1](#), these velocities fall well within the optimal range for effective debris transport.

Table 1—Annular velocity outcomes for various fluid strategies.

Fluid Type	Pump Rate, bbl/min	Nitrogen Rate, scf/min	Annular Velocity 7-in. section, ft/min	Annular Velocity 9 5/8-in. section, ft/min	Comments
Lean Newtonian Fluid	2.5	0	~80	~35	Suboptimal velocities; ineffective transport
Nitrified Newtonian Fluid	2.5	800	~125–170	~80–190	Optimized for debris lifting and foam stability
Nitrified Gelled Fluid	2.5	500	>60	>60	Meets recommended gel-fluid threshold for solids transport

By selecting a nitrified fluid system utilizing a Newtonian base fluid in combination with periodic batches of gelled fluid, the annular velocity challenge was effectively addressed. This approach ensured adequate hole cleaning performance across both casing sections. However, two critical technical gaps remained.

The first challenge was the selection of a stable gel system compatible with the nitrified fluid environment. Stability under multiphase flow conditions was essential to maintain the rheological properties required for effective solids transport and suspension. The second challenge involved identifying a suitable downhole jetting tool capable of delivering sufficient hydraulic energy to effectively clean the inner walls of the 7-in. liner. Ensuring adequate jetting force was crucial for dislodging adhered debris and facilitating its removal through the annular flow path.

A high viscous (HV) gel was selected as the primary cleanout fluid for this operation due to its ability to maintain consistently high viscosity under a wide range of shear conditions and temperatures ([Fig. 2](#)). This rheological stability enhances the carrying capacity of the gel, making it particularly effective in suspending and transporting solid debris—such as scale, sand, or residual plug materials—during this CT cleanout in both single or multi-phase cleanout regimes. Unlike conventional brines or low-viscosity fluids, this gel resists thinning under dynamic flow regimes, ensuring efficient hole cleaning across the big bore section.

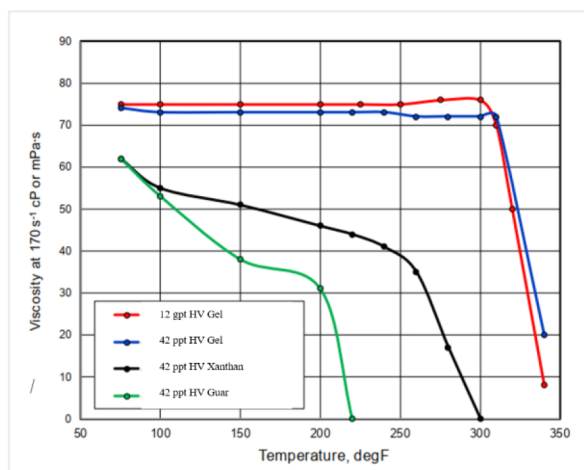


Figure 2—Rheology of various well clean out fluids.

Rotary High-Pressure Jetting Tool

Engineered high-pressure jetting service is pivotal in the descaling of scale deposits (Fig. 3). The tool's controlled swivel rotation ensures comprehensive 360-degree wellbore coverage where key design features include carbide nozzles, encapsulated, and lubricated heavy-duty bearings, and a maximum operating temperature of 350°F (177°C), enabling the tool to withstand harsh chemical environments and abrasive fluids. The high-velocity fluid stream, essential for effective downhole jetting, minimizes energy dissipation in the tool itself, ensuring efficient scale removal and reducing treatment time and chemical usage. By focusing fluid energy at the jet interface and ensuring coherent high-velocity streams, the tool achieves superior downhole performance, making it indispensable for effective clean out. (Alaskar, S. Y, 2024).



Figure 3—Rotary High-Pressure Jetting Tool.

In this specific operation, the rotary high-pressure jetting tool played a critical role in restoring access to the top of the stuck mechanical plug, which was entirely covered by consolidated sulfate scale. The tool's 360-degree coverage allowed for thorough descaling of the inner walls of the 7-in. liner, including eccentric and angled sections near the liner hanger. The high-impact jetting capability effectively dislodged both soft and hard debris types, as evidenced by the recovery of 154 kg of solids at surface—including 134 kg of soft sulfate scale and 20 kg of metallic fragments. Tool performance was further validated by clean returns and successful tag of the top of fish (TOF), confirming that debris had been sufficiently cleared without damaging the plug. The use of this tool not only minimized the need for additional mechanical intervention but also ensured efficient scale removal in a single cleanout run, thereby reducing total operational time and enhancing the success probability of the subsequent fishing operation.

Coiled Tubing Size Selection

Tubing force modeling (Table 2) was performed to evaluate different pipe sizes and ensure sufficient overpull capacity throughout the operation. The analysis was instrumental in selecting an appropriate tubing configuration that would provide the mechanical reliability required to safely and effectively execute the operation, thereby increasing the overall confidence of the operator in the proposed solution.

Table 2—Summary of tubing forces modeling.

CT OD Size, in.	Max Pulling Force, lb	Maximum Slackoff, lb	Maximum Pump Rate, bbl/min	Effective Cycling
2	33,669	27,601	2.1	337
2 3/8	48,037	29,264	2.6	132
2 7/8	57,825	31,389	3.0	68

A 2 3/8-in. 90ksi yield strength CT was selected to provide optimum pumping rates for effective clean out at the top of the plug, as well as to achieve the targeted overpull for successful plug retrieval.

Fishing Tool

A 6-in. GS flow-release pulling tool (Fig. 4) was used to selectively latch/unlatch internally into the plug profile, apply overpull to retract the plug, and pull out of hole to surface. The tool is activated by differential pressure generated through pumping, which retracts the keys and allows the tool to enter the fishneck profile. Once pumping is stopped, the tool deactivates, allowing the keys to extend and securely latch into the profile.

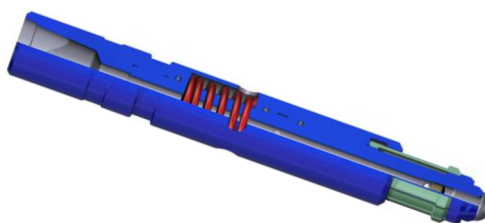


Figure 4—GS flow-release pulling tool.

Wellhead Stack and Centralization

Deploying a 6-in. OD CT bottomhole assembly (BHA) is atypical and necessitates the use of a specialized 7 1/16-in. pressure control equipment (PCE) composed by blowout preventer (BOP) and riser system to accommodate both the oversized BHA and the associated fishing tools. This configuration introduces unique surface equipment challenges, particularly with respect to tool clearance and safe deployment.

To ensure successful passage through the wellhead components and critical internal restrictions—such as the transition from the 9 5/8-in. casing to the 7-in. liner via the liner hanger—precise BHA centralization was essential. Proper centralization mitigates the risk of mechanical interference, ensures smooth navigation through the wellhead "jewelry," and enhances overall operational reliability. This setup was rigorously reviewed to meet both mechanical integrity and operational efficiency requirements under the aforementioned well conditions.

Operation Flowchart

Integrating all of the above solutions into a decision-making flowchart (Fig. 5) was essential for the smooth planning and execution of the intervention.

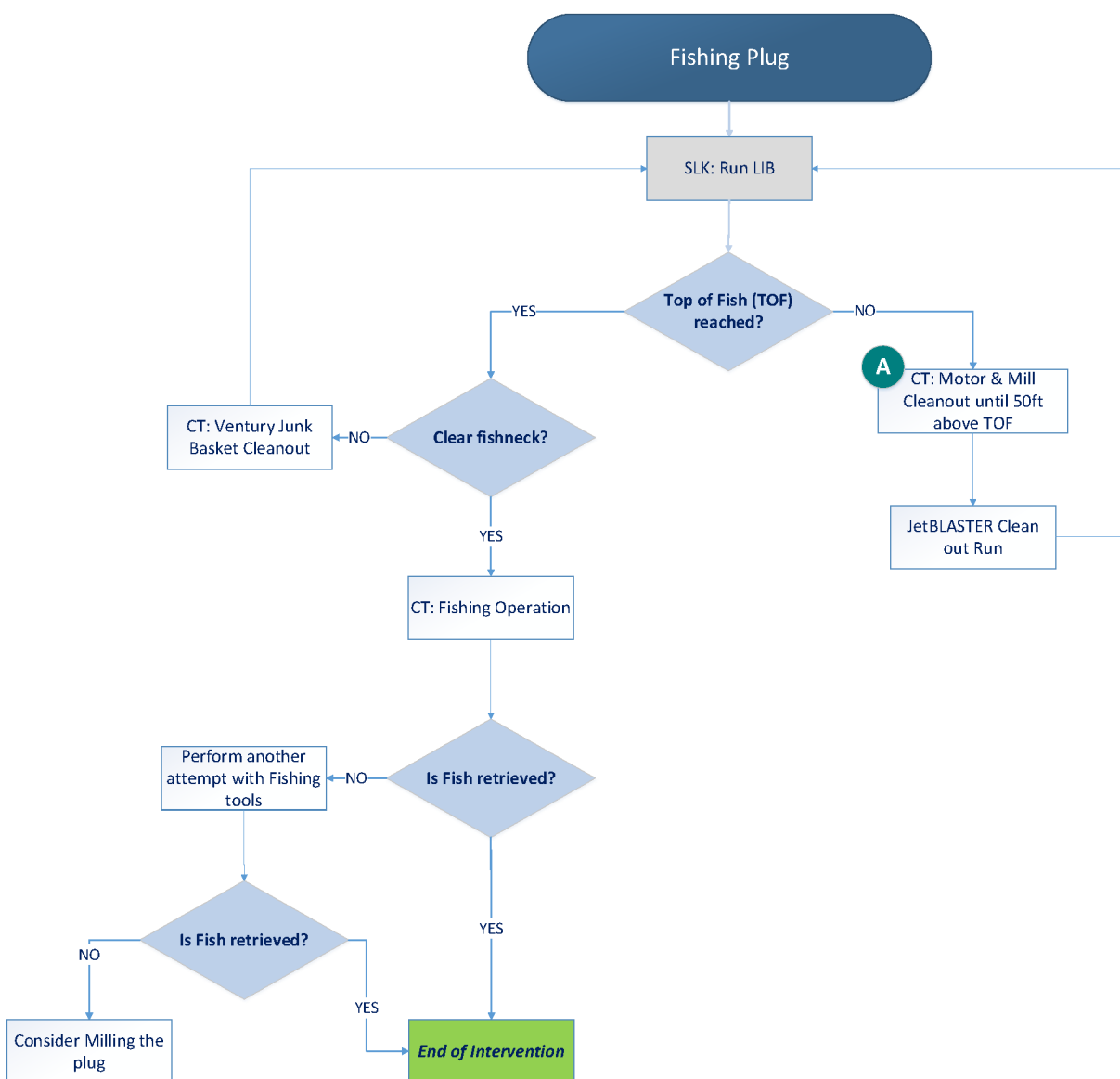


Figure 5—Decision-making flowchart for the CT intervention.

Risk Assessment and Contingency Planning

Given the inherent uncertainties and complexity of fishing operations, extensive pre-job planning was undertaken to evaluate all potential scenarios. A comprehensive risk assessment and contingency planning process was carried out, addressing mechanical, operational, and downhole risks. While not all identified scenarios occurred during execution, this thorough preparation was critical to ensuring operational flexibility and minimizing non-productive time. The planning phase involved detailed discussions, cross-functional reviews, and scenario-based action plans to proactively address possible challenges, highlighting the importance of robust contingency design in fishing intervention.

The following scenarios outline the key technical risks identified during the planning phase, along with the corresponding mitigation strategies deployed during the operation. Each scenario was evaluated for potential impact on job success and addressed through customized contingency plans.

Scenario 1: Unable to Establish Suitable Fluid Circulation

Key considerations

- The 9 5/8-in. casing transitioning into a 7-in. liner, combined with the 2 3/8-in. CT, results in suboptimal annular velocities. This limits the ability to effectively lift debris to the surface, which is crucial for achieving successful circulation.
- [Table 1](#) consolidates the annular velocity outcomes for different fluid strategies, highlighting the advantages of nitrified systems in overcoming circulation challenges.

Contingency

- Perform clean out all the way with HV gel fluid for effective lifting of debris. See [Fig. 2](#).
- To address insufficient circulation, nitrogen will be introduced to ensure suitable lifting and improved debris transport to the surface.
- Multiple gel sweeps should be performed and circulated to ensure the return of debris in a timely manner, while monitoring returns closely.

Scenario 2: Inability to Clean Out Debris with Rotary High-Pressure Jetting Nozzle

Key considerations

- Understand the nature and type of debris.
- If the debris are particularly hard or dense, the rotary high-pressure jetting nozzle may not be able to clear it effectively.

Potential Contingency

- A dedicated venturi junk basket run to remove any remaining loose debris.
- A lead impression block (LIB) run to verify the cleanliness and geometry at TOF.
- If the jetting tool fails to clear the hard obstruction, an additional run with milling BHA shall be considered.
- A small volume of custom-fit scale dissolver (e.g., 10–15 bbl) could be spotted above the anticipated TOF to chemically soften the debris.

Scenario 3: Debris Accumulation on Top of Fish

Key considerations

- Debris settled and compacted above TOF.
- Debris or large particles may settle on TOF during the clean-out operation within the 7-in. liner, particularly if annular velocities are insufficient to lift the debris to the surface.
- The settled debris may obstruct further operations, complicating the fishing process.

Potential Contingency

- To address debris accumulation, a venturi junk basket run should be executed.

- The recovery of debris should be checked at the surface. If the venturi is full, additional runs may be required to remove all debris.
- As an additional step, 10 bbl of scale dissolver shall be spotted with CT to chemically dissolve any remaining debris, ensuring a more thorough cleanout.
- After the scale dissolver treatment, circulate out remaining debris with gel sweeps to verify that the TOF is clean and conditioned for further fishing attempts.

Scenario 4: CT Unable to Latch the Fish

Key considerations

- Despite using a 5.80-in. stabilizer, there is a possibility that due to buckling or bending of the 2 3/8-in. CT, the BHA may not be able to latch onto the fish properly. See [Fig. 6](#) and [Table 3](#) for details of the fishing BHA.

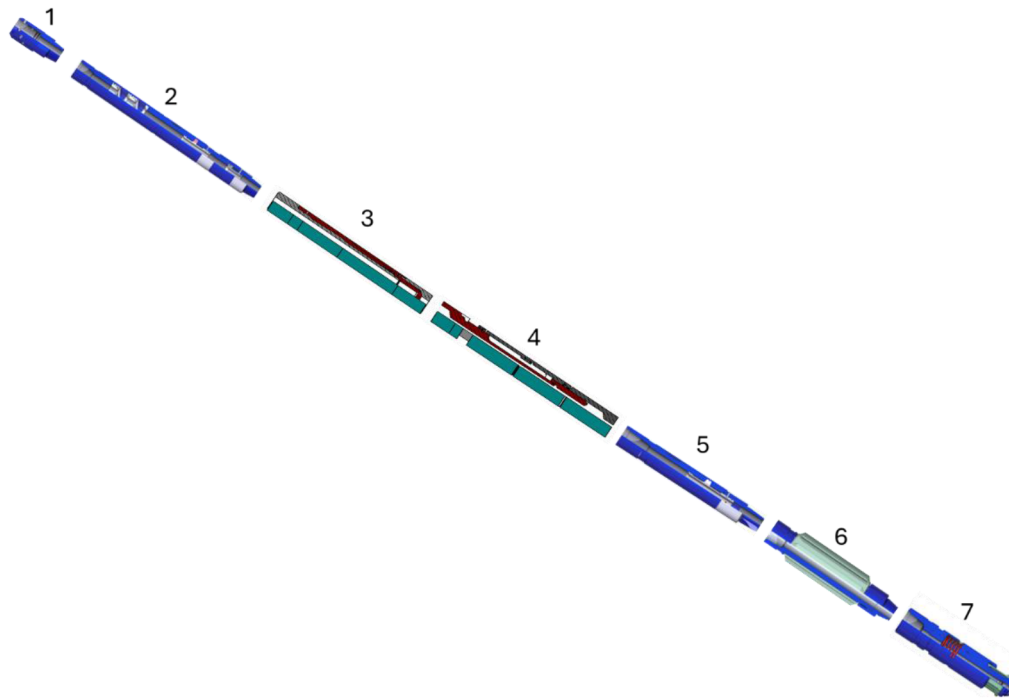


Figure 6—Representation of the Fishing BHA schematic.

Table 3—Description of Fishing BHA components.

Item	Description	OD, in.	Length, ft
1	External Connector	2 7/8	0.71
2	Rugged MHA	2 7/8	2.94
4	Accelerator	2 7/8	8.67
5	Jar	2 7/8	7.00
6	Lower Disconnect	2 7/8	1.13
6	Stabilizer	5.800	1.75
7	6.0-in. GS Spear Flow Release	5 1/2	1.33

Potential Contingency

- Multiple latching attempts should be performed by adjusting the CT depth incrementally (~ 2–3 ft) around the expected fish neck position, combined with selective pumping to activate the internal GS pulling tool. Each attempt includes pumping at low rates (e.g., 0.5–0.7 bbl/min) to create the differential pressure needed to retract the tool's keys, followed by cessation of pumping to allow the keys to expand and attempt engagement. If no positive indication of latch is observed (e.g., pressure increase or weight change), repositioning should be repeated. If latching continues to fail, pull out to inspect the BHA and consider adding stabilizers or adjusting the tool string configuration for better alignment.
- If necessary, perform a shorter LIB run with a larger stabilizer to confirm the path is clear and the BHA can properly reach the fish profile.

Scenario 5: CT Latch the Fish but Cannot Retrieve the Fish to Surface

Key considerations

- Fish may be stuck from slips or element area.
- Debris compacted on the top of fish and CT received overpull after latching.
- [Table 4](#) summarizes the force thresholds and mechanical limits encountered during the fishing operation, outlining the safe operating envelope for plug retrieval.

Table 4—Force Thresholds and Overpull Limits for Plug Retrieval.

Condition / Scenario	Force / Overpull lbf	Remarks
Equalizing pressure using pump-thru plug	7,800	Flapper valve opens to balance pressure
Required force to retrieve plug (normal conditions)	10,400	Standard pull force to unset the plug
Maximum safe overpull if stuck on slips	34,000	Above this, slip springs may begin to break
Maximum overpull by CT (job-dependent)	45,000–50,000	Mechanical limit of CT string during job
Safe applied overpull if stuck at element area	60,000	Upper safe threshold before internal damage risk
Theoretical failure value of plug (fish neck breaks free)	132,000	Plug internals break, neck detaches from body

Potential Contingency

- Perform over pull until 34,000 lb; if there is no progress, release the over pull and try again.
- Use jar to hammer the fish upward within CT limits.

Scenario 6: Excessive Dragging While Retrieving the Fish to Surface

Key considerations

- When pulling the fish, drag may occur due to deformation of the packer element, even after deflation of the plug, or due to debris accumulation within the wellbore or liner around the packer.
- The dragging could lead to overpull during the pull-out operation.

Potential Contingency

- As an intermediate contingency, initiate annular pumping through the kill port using treated water or base fluid at 1.0–1.5 bbl/min for 5–10 minutes (approximately 5–15 bbl total volume), depending on well depth and observed drag. This helps dislodge any accumulated debris around the plug or BHA by flushing it upward and reducing friction during pull-out. The rate should be high enough to create sufficient annular velocity without exceeding surface pressure limits or compromising BHA stability.
- If the dragging is within the acceptable limits of the CT and the overpull does not exceed 10,000 lb, continue with a controlled pull-out. Ensure the pulling speed remains slow and steady to avoid increasing drag or damaging the CT. Monitor weight and tension limits carefully throughout the operation.
- If drag begins to approach the maximum limit, stop the CT movement, release the fish using the release mechanism or confirm discuss with engineering if possibility to push the fish downhole using the new BHA

Scenario 7: CT tag at the top of liner or at the Tubing Hanger

Key considerations

- CT tag the top of liner due to change in ID from 8.835-in. to 6.276-in.
- Ensure slack off remains within the CT and BHA limits. If slack off exceeds the limit, pull out of hole (POOH) and inspect the CT pipe and BHA.

Potential Contingency

- Use stabilizer and straight bar in BHA.
- When making the connector make sure the CT is as straight as possible. If needed, trim some extra CT pipe section.
- During operation, pick up CT to change orientation of BHA and slowly run in hole (RIH) to pass tag depth. Try pumping at various rates to pass.

Coiled Tubing Intervention

A two-run CT intervention was executed to restore access and retrieve a stuck mechanical plug set inside a 7-in. liner. The well geometry and scale accumulation presented considerable risks, making precise intervention planning and execution essential. The operation demonstrated the versatility of CT in executing complex interventions in high-risk environments without the need for a workover rig.

Run 1: Cleanout Operation

The initial run aimed to remove sulfate scale and debris accumulation, above the mechanical plug using a high-pressure CT cleanout assembly equipped with a rotary high-pressure jetting nozzle and a 4.25-in. drift ring. It was inferred from a prior slickline LIB run that tagged near 4,000 ft—significantly shallower than the expected plug depth—and showed no impression marks, suggesting a layer of hard scale or compacted debris was preventing direct contact with the plug. A 2 3/8-in. high-pressure CT string was deployed while maintaining pull tests every 1,000–1,500 ft. Surface circulation was established using a blend of treated water and high-viscosity gel, with annular velocities modeled to exceed minimum transport requirements.

The cleanout operation (Fig. 7) progressed with a reciprocation strategy and staged bottoms-up sweeps. Circulation was optimized using nitrified fluids to enhance lifting efficiency in the large ID liner. Surface

testing confirmed solid recovery of sulfate scales. A total of 154 kg of debris was recovered during this run, including significant deposits of solid scale. A hard tag was observed close to 4,115 ft with a 2,000-lbf slack off, confirming the plug depth. After clean returns were observed, the CT was pulled out of hole to prepare for the fishing operation. Table 5 provides a description of the operational events highlighted in Fig. 7.

Table 5—Description of the events from CT cleanout run.

Event	Description
1	CT RIH to 3,000 ft while performing periodic pull tests every 1,000-ft and pumping rate of 1.0 bbl/min with choke size 32/64-in.
2	CT RIH to 3,800 ft while fluid rate adjusted to 2.5 bbl/min and nitrogen rate to 500 scf/min with choke size 52/64-in.
3	From 3,800 ft till 4,115 ft, performed the cleanout in nitrified mode while pumping 25 bbl HV gel after 50 ft penetration and sweep.
4	Once received clean return with zero recovery, CT will POOH to surface for next run while pumping at minimum rate.

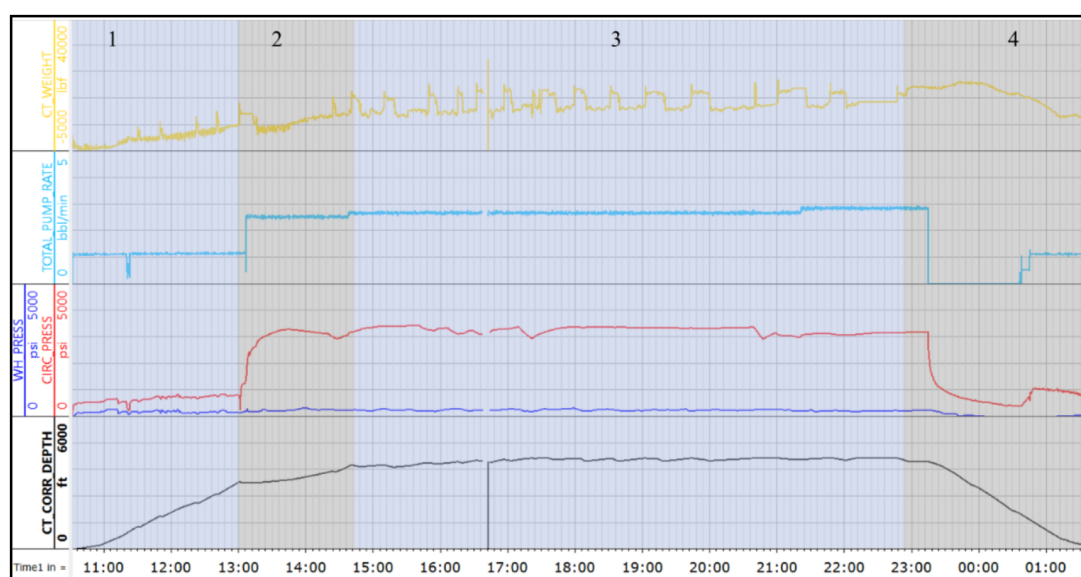


Figure 7—Acquisition plot for CT cleanout operation. From top to bottom: CT weight (yellow), pump rate (light blue), CT Circulation pressure (red), wellhead pressure (blue), CT depth (black).

The cleanout operation was conducted, involving the pumping of gel after 50 ft bite was penetrated and a sweep was completed. During reciprocation, parameters such as wellhead pressure, circulation pressure, pump rate, nitrogen rate, and choke size were adjusted as needed. The cleanout was completed up to the target depth, and a tag was detected, leading to the CT pipe being flagged at the plug depth. Fig. 8 shows the zoomed plot for the clean out operation.

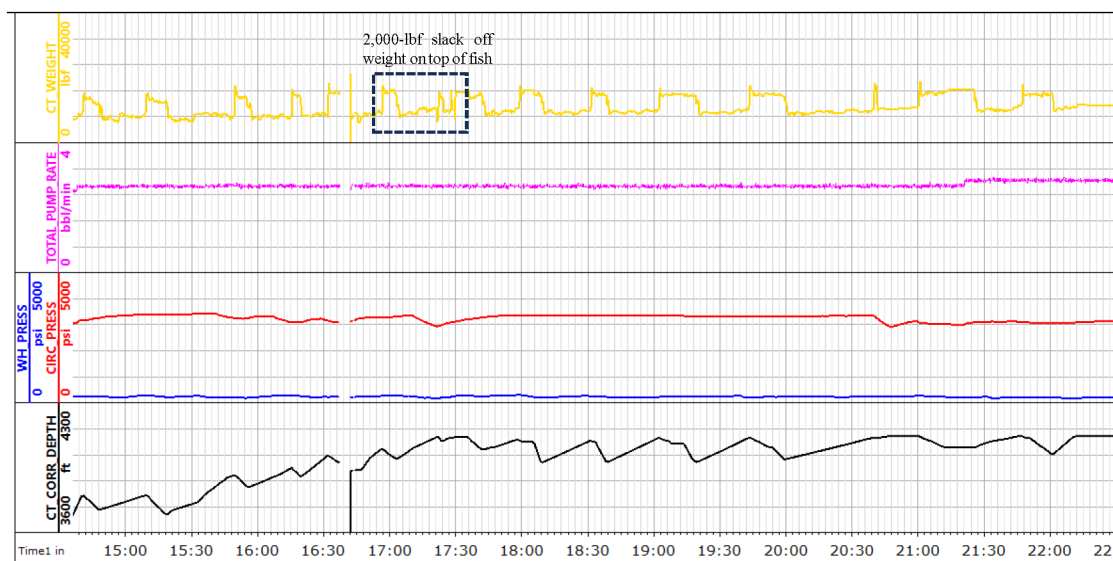


Figure 8—Zoomed plot for CT Cleanout operation. From top to bottom: CT weight (yellow), pump rate (light blue), CT Circulation pressure (red), wellhead pressure (blue), CT depth (black).

Debris characterization was conducted through a basic solubility test sequence. The soft debris (Fig. 9) neither float nor dissolve in water, but dissolved readily in hydrochloric acid while releasing a rotten egg odor, indicative of H_2S . This suggested the debris was primarily iron sulfide (FeS) scale, a common byproduct of corrosion in water injector wells with sulfide-rich environments. The 134 kg of FeS was likely formed over extended shut-in periods or during the well killing phase.



Figure 9—Sample of the soft debris recovered during the cleanout operation.

The harder metallic debris (Fig. 10), totaling 20 kg, did not exhibit similar reactivity. While its exact origin remains uncertain, it is suspected to be from either previous intervention attempts, erosion or wear of wellbore components, or residual fragments from earlier downhole tool deployments.



Figure 10—Sample of the hard debris recovered during cleanout run.

Run 2: Fishing Operation

The second run involved fishing the stuck mechanical plug using a 6-in. internal GS pulling tool conveyed on the same 2 3/8-in. CT string. Function testing and tool readiness were confirmed prior to RIH. The GS tool was activated by pumping at 0.7 bbl/min, with downhole pressure response indicating successful engagement.

CT latched onto the plug while pumping at 0.7 bbl/min, with a 200-psi increase in circulation pressure confirming engagement (Fig 11). Pumping was then stopped to allow the pulling tool keys to fully expand into the fish neck profile. After waiting around 5 minutes, the CT was gently picked up to confirm secure latch. The reel manifold valve was then closed, and overpull gradually applied. A maximum of 34,000 lb. was reached, which successfully unseated the plug from its set position. The CT was then pulled up 10 ft and held for 30 minutes to allow full retraction of the plug elements. Real-time acquisition data, including tubing force monitoring and weight indicators, validated fishing success without exceeding mechanical limitations. Fig. 12 shows the mechanical plug retrieved at surface.

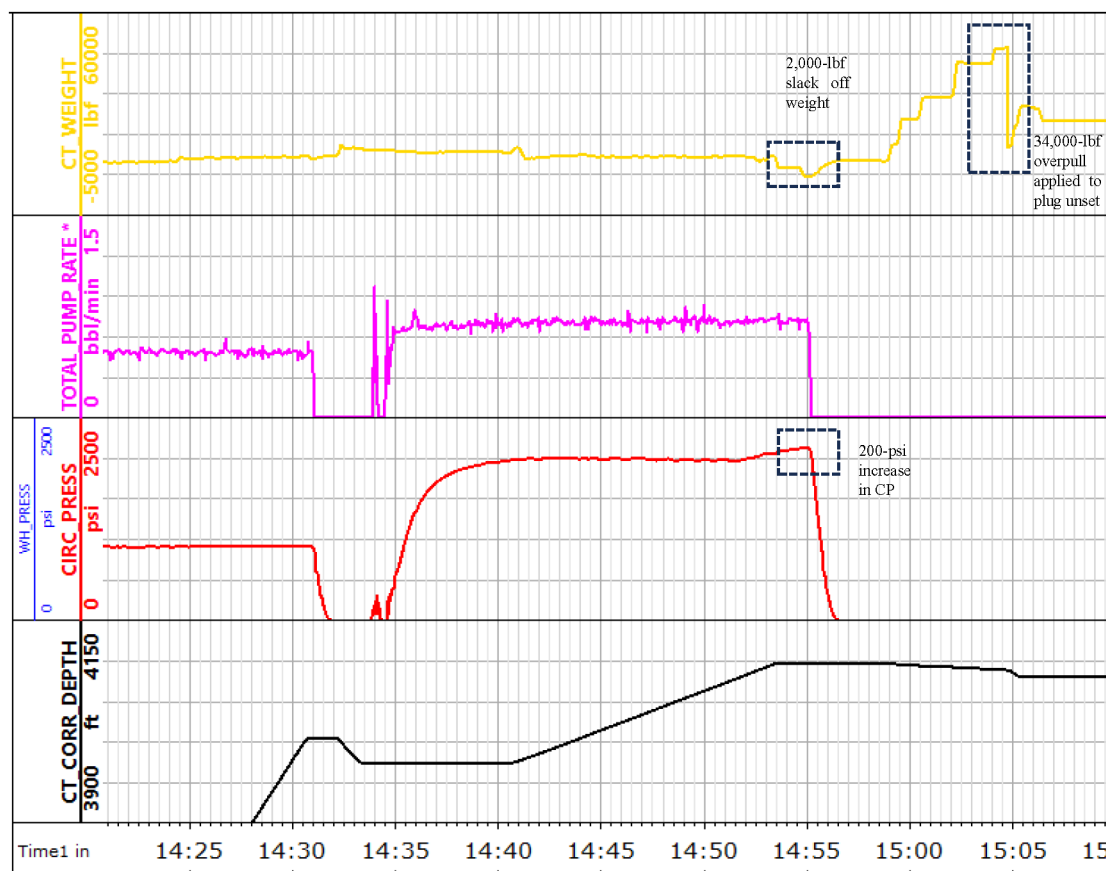


Figure 11—Snapshot of acquisition plot for CT fishing operation. From top to bottom: CT weight (yellow), pump rate (magenta), CT Circulation pressure (blue), wellhead pressure (blue), CT depth (black).

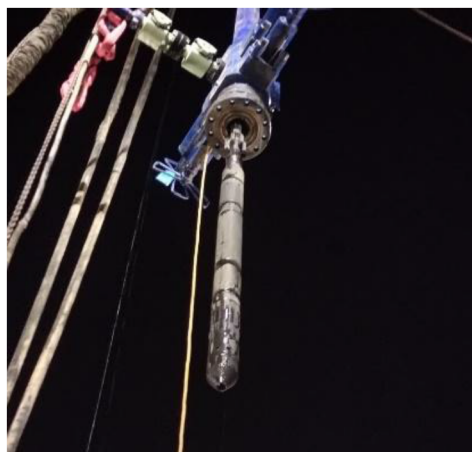


Figure 12—Picture of the retrieved fish at surface.

Results

The outcomes of these interventions underscore the efficacy of CT in complex well scenarios.

Operational Outcomes

- Cleanout run: Utilizing a rotary high-pressure jetting nozzle with a 4.25-in. drift ring, the cleanout operation successfully removed 154 kg of sulfate scale and debris. The operation achieved a maximum depth of 4,115 ft, ensuring unobstructed access to the mechanical plug.

- Fishing run: The subsequent fishing operation employed a 6-in. internal GS pulling tool, effectively retrieving the mechanical plug. Controlled overpulls up to 34,000 lbf were applied without exceeding mechanical limitations, confirming the robustness of the CT string and tool assembly.

Efficiency and Time Savings

- Compared to traditional rig-based interventions, the CT approach demonstrated significant time and cost efficiencies:
- Reduced Downtime: The entire intervention was completed in approximately 2.5 days, minimizing production downtime.
- Cost-Effectiveness: Eliminating the need for a workover rig resulted in substantial cost savings, including reduced mobilization expenses and personnel requirements.
- Operational Flexibility: The CT intervention allowed real-time adjustments based on downhole conditions, enhancing operational responsiveness.

Safety and Environmental Considerations

- The intervention was conducted with zero non-productive time (NPT) and no safety incidents, highlighting the safety advantages of CT operations, especially in high-risk environments.

A rig-based fishing operation was initially considered as a contingency for retrieving the stuck mechanical plug; however, the CT intervention provided a more efficient and lower-risk alternative for this particular well profile. Rig-based fishing would have required complete rig mobilization, pressure control rig-up, and significant non-productive time (NPT) due to complex wellsite logistics, all contributing to an estimated 7–10 days of operational time. In contrast, CT enabled precise, depth-controlled access to the target zone with significantly reduced surface footprint and operational complexity. The use of CT allowed for immediate cleanout of debris using tailored fluid systems and high annular velocity, followed seamlessly by a fishing run without rigging down. Overall, the CT approach reduced job duration by over 60%, minimized deferred production, and eliminated the need for heavy lifting operations, making it a safer and more cost-effective solution compared to traditional rig-based methods. [Table 6](#) shows Comparative Analysis: CT vs. rig-based intervention.

Table 6—Comparative analysis: CT vs. rig-based Intervention.

Parameter	Coiled Tubing Intervention	Rig-Based Intervention
Total Duration	~2.5 days	~7–10 days
Mobilization Requirements	Minimal	Extensive
Operational Flexibility	High	Moderate
Cost Implications	Lower	Higher
Safety Risks	Lower	Higher
Environmental Impact	Minimal	Significant

Conclusions

Collaboration between operator and CT service provider was essential for the successful cleanout and fishing operation, in a working environment where workover rig had been the preferred intervention method.

The extensive CT modeling, as well as the cleanout elements selected for this operation, such as HV gel, nitrified fluids, rotary high-pressure jetting tool, CT size, pumping rates, were instrumental for recovering 134 kg of soft debris and 20 kg of hard metallic debris across a large size completion (9 5/8-in. × 7-in.) and let the TOF clear for the fishing operation.

The fishing operation was smoothly executed. In 2.5 days intervention, the CT intervention showed the main benefit by minimizing production downtime, averting the need for a workover rig resulted in substantial cost savings.

As the industry strives for cost-effective and sustainable interventions, the case study presented in this paper provides an example of how a custom-fit CT intervention workflow for complex fishing operation saved a costly workover rig intervention, while reducing near 70% CO₂ emissions.

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